# Flatcar Container Linux

**Minimal. Immutable. Secure by default.**

A **fully free and open-source, community-driven** Linux distribution that is minimal, security-focused and **optimized to run containers**.
A **CNCF project** that is always up to date and gives you full control over your fleet.

## Why Flatcar

### Immutable, image-based

A minimal footprint with a read-only, verified `/usr`. No package manager on the node, so every machine is the same.

### Self-updating, with rollback

A/B partitions apply updates in the background. You control when to reboot into the new image, and a bad update rolls back automatically.

### Operating System as Code

Describe the machine in YAML and Ignition applies units, files, SSH keys, users and disks at first boot. No interactive setup.

### Extensible with sysexts

Add Docker, Podman, Kubernetes or GPU drivers as systemd-sysext overlays, pinned to the version you want.

**Supported wherever you run your containers:** every major cloud, on-prem and bare metal

## Get started and join us

**Website**
flatcar.org

**GitHub**
flatcar/Flatcar

**Learning series**
learn.flatcar.org

**Discord**
chat.flatcar.org

COME SAY HI

**Join our Discord server** to ask questions and chat with the maintainers: chat.flatcar.org
Office hours and dev syncs: every 2nd and 4th Thursday, 2:30 pm UTC

## How Flatcar differs

| TRADITIONAL SERVER OS | FLATCAR |
| --- | --- |
| Anyone can change software at any time | Read-only system; extras run as containers or system extensions |
| Patches packages one by one, in place | Writes a whole new image to a second partition |
| A failed update can leave a server broken | A failed boot rolls back automatically |
| Server configuration slowly drifts apart | Every node stays identical, defined as code and set up at first boot |

## Trusted in production

### Run at scale

Companies behind products you may already use rely on Flatcar across fleets of tens of thousands of nodes.

### Adopters

Adobe, STACKIT, DeepL, AT&T, 1&1 Mail & Media, Equinix Metal, Qualys, Wipro and many more.

### GPU and AI ready

The NVIDIA driver is its own system extension, so the OS and the driver update separately.

### Vendor neutral

A CNCF project guided by Steering and Technical committees, which keeps the project neutral and open.

## Frequently Asked Questions

### What does immutable mean?

The core of the operating system is read-only, while your data and configuration stay writable. Updates replace the OS image, so all machines stay identical.

### What does declarative mean?

Think of it as Operating System as Code, like a Dockerfile for your OS. You describe the machine you want in a short YAML file, and it is applied automatically at first boot.

### Can I install software on it?

Yes, just not with a package manager on the host. Run your apps as containers, and add extra host tools as system extensions: only the runtime pieces you need.

### What can I run on it?

Any container workload, with Docker, containerd, Podman or whatever runtime you prefer. Kubernetes is popular but optional, and you can add your own tools as system extensions.

### How often does it update?

From every couple of weeks to every couple of years. Pick a channel (Alpha, Beta, Stable or LTS), then leave updates automatic or control them yourself.

### How do I get started?

Run it locally in QEMU with the Learning Series at learn.flatcar.org, no cloud account needed.